THE UNIVERSITY of NORTH CAROLINA at CHAPEL HILL

Boundary-Aware Redirected Walking for Virtual Heritage Exploration in Constrained Spaces

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Abstract

Redirected walking (RDW) allows for users to explore large VR environments while being in small physical spaces through subtly manipulating their perceived trajectory. This adjustment occurs through rotation, curvature, and translation gains. This project uses a RDW system so that participants can experience walking around a virtual reconstruction of the Swayambhunath temple complex in Kathmandu, Nepal, within a physical space as small as 4 square meters.

Above: A photograph of the Swayambhunath temple complex.

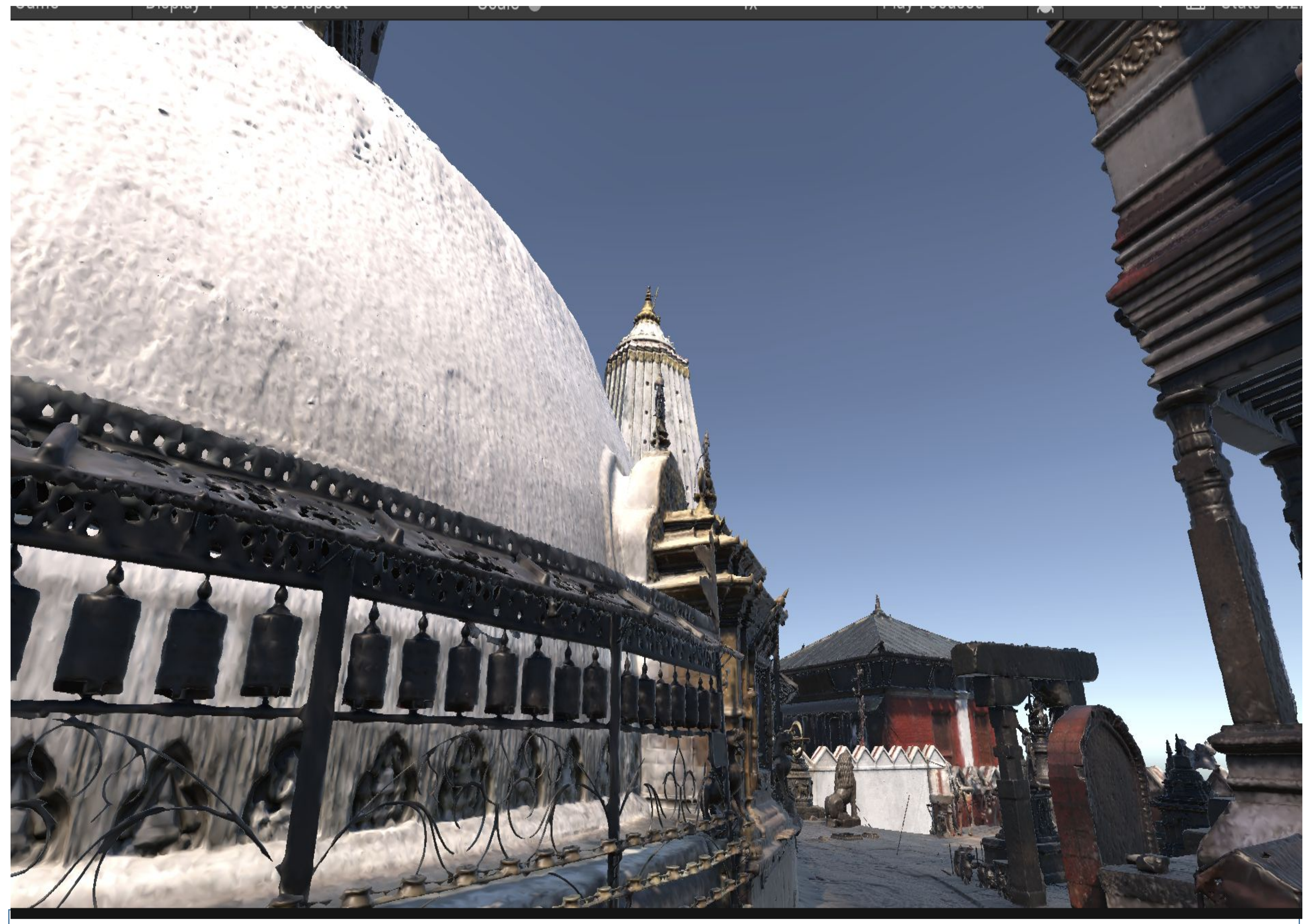
Background

Through our collaboration with the Religious Studies Department, we have exclusive access to 3D scan the Swayambhu Mahachaitya in Nepal. We wanted to create an experience in which students can experience the temple from anywhere in the world and walk around it on an approximate 1-to-1 scale. This project aims to make the experience and study of religious heritage sites such as this more accessible.

Methods

Our experience runs in both Unity and Unreal Engine on the Meta Quest 3 headset. We utilize the internal tracking of the Meta Quest to find the users position. The experience calculates the nearest points on the circle based on the user's trajectory, then rotates the horizon of the world to guide them away from physical environmental hazards.
We used translation gains, rotation gains, and curvature gains to guide participants as they follow a curved virtual path.

Left: A participant using the Meta Quest 3 headset is directed to walk around the central stupa of the virtual temple model.



Above: A screencapture of the Swayambhu temple complex model.

Results

With 3-4 revolutions of the 2m wide space, a scaled model of the 50m temple complex was successfully traversed. Space-specific calibration resulted in minimal discomfort for participants and constrained their paths within the safe radius of the space. If participants did not follow the directed path, a virtual boundary wall would warn them before they crossed out of the space.

Similar immersive cultural experiences can be hosted in a confined space using the techniques practiced.

Above: An aerial view of the virtual temple complex. The highlighted circle depicts the path participants were instructed to take.

References

- M. Rietzler, J. Gugenheimer, T. Hirzle, M. Deubzer, E. Langbehn and E. Rukzio, "Rethinking Redirected Walking: On the Use of Curvature Gains Beyond Perceptual Limitations and Revisiting Bending Gains," 2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Munich, Germany, 2018, pp. 115-122, doi: 10.1109/ISMAR.2018.00041. keywords: {Legged locomotion;Measurement;Meters;Navigation;Virtual environments;Teleportation;Augmented reality},
- J. -J. Chen, H. -C. Hung, Y. -R. Sun and J. -H. Chuang, "APF-S2T: Steering to Target Redirection Walking Based on Artificial Potential Fields," in IEEE Transactions on Visualization and Computer Graphics, vol. 30, no. 5, pp. 2464-2473, May 2024, doi: 10.1109/TVCG.2024.3372052. keywords: {Legged locomotion;Aerospace electronics;Virtual environments;Force;Vectors;Layout;Robots;Redirected walking;artificial potential field;locomotion;virtual reality},
- N. L. Williams and T. C. Peck, "Estimation of Rotation Gain Thresholds Considering FOV, Gender, and Distractors," in IEEE Transactions on Visualization and Computer Graphics, vol. 25, no. 11, pp. 3158-3168, Nov. 2019, doi: 10.1109/TVCG.2019.2932213.
- keywords: {Visualization;Legged locomotion;Observers;Usability;Virtual environments;Stimulated emission;Virtual reality;Locomotion;Perception;Detection thresholds;Distractors;Gender differences;Simulator sickness},